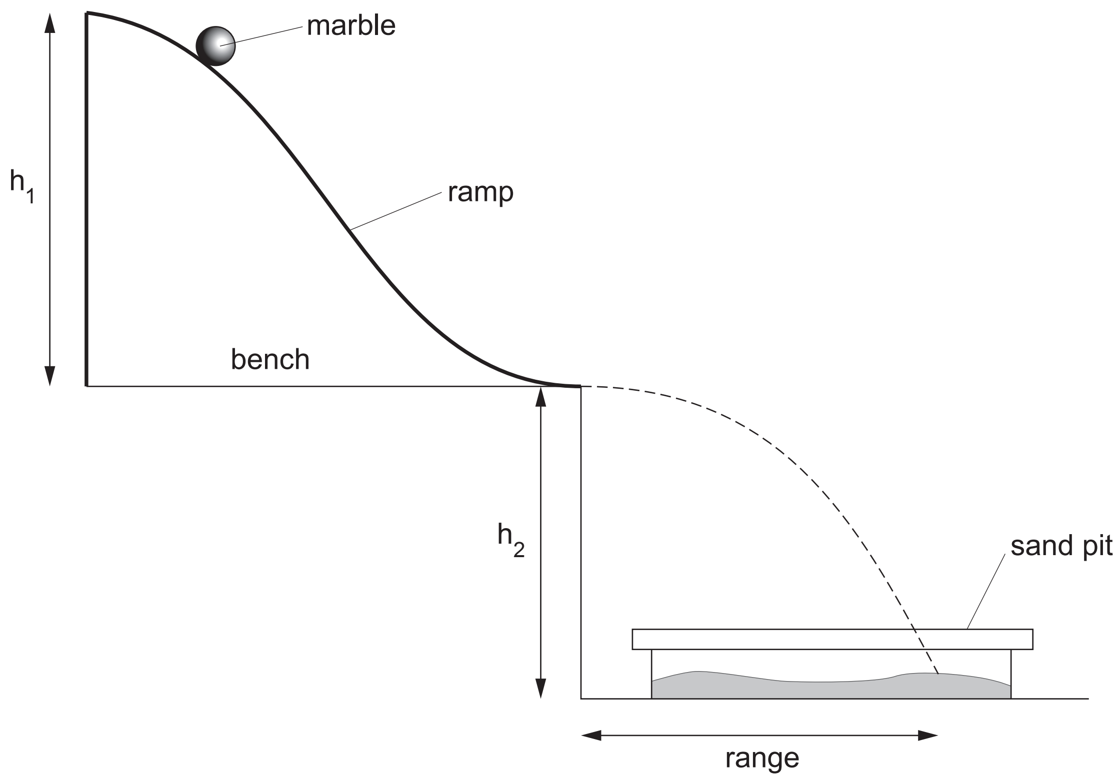
Examiner only

6. A model ski jump can be set up on a bench as shown in the diagram below.



The marble is released from rest at the top of the ramp. It leaves the bottom of the ramp with a certain velocity. It moves through the air and lands in the sand pit. The horizontal distance travelled in the air is called the **range**.

(a) (i) For one height (h_1) of the ramp, the mean range was measured as 45 cm and the time the marble was in the air was 0.5 s.

Use the equation:

$$\text{velocity} = \frac{\text{mean range}}{\text{time to travel the range}}$$

to calculate the velocity of the marble as it left the ramp. [2]

Velocity = ... cm/s

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(ii) The time for the marble to run down the ramp was measured 5 times. The times were 1.2 s, 1.3 s, 1.7 s, 1.2 s and 1.1 s.

I. State which time was an anomalous result. .... [1]

II. Calculate the mean time for the marble to travel down the ramp. [2]

Mean time to travel down ramp = ... s

(iii) Use your answers from (i) and (ii) and the equation:

$$\text{acceleration} = \frac{\text{velocity of marble as it left the ramp}}{\text{mean time to travel down the ramp}}$$

to calculate the acceleration of the marble down the ramp. [2]

Acceleration = ... cm/s^2

(b) Another experiment was carried out to find out how the acceleration of the marble depended on the height h_1 of the ramp.

(i) The same marble was used each time. State **two** other variables that should be controlled. [2]

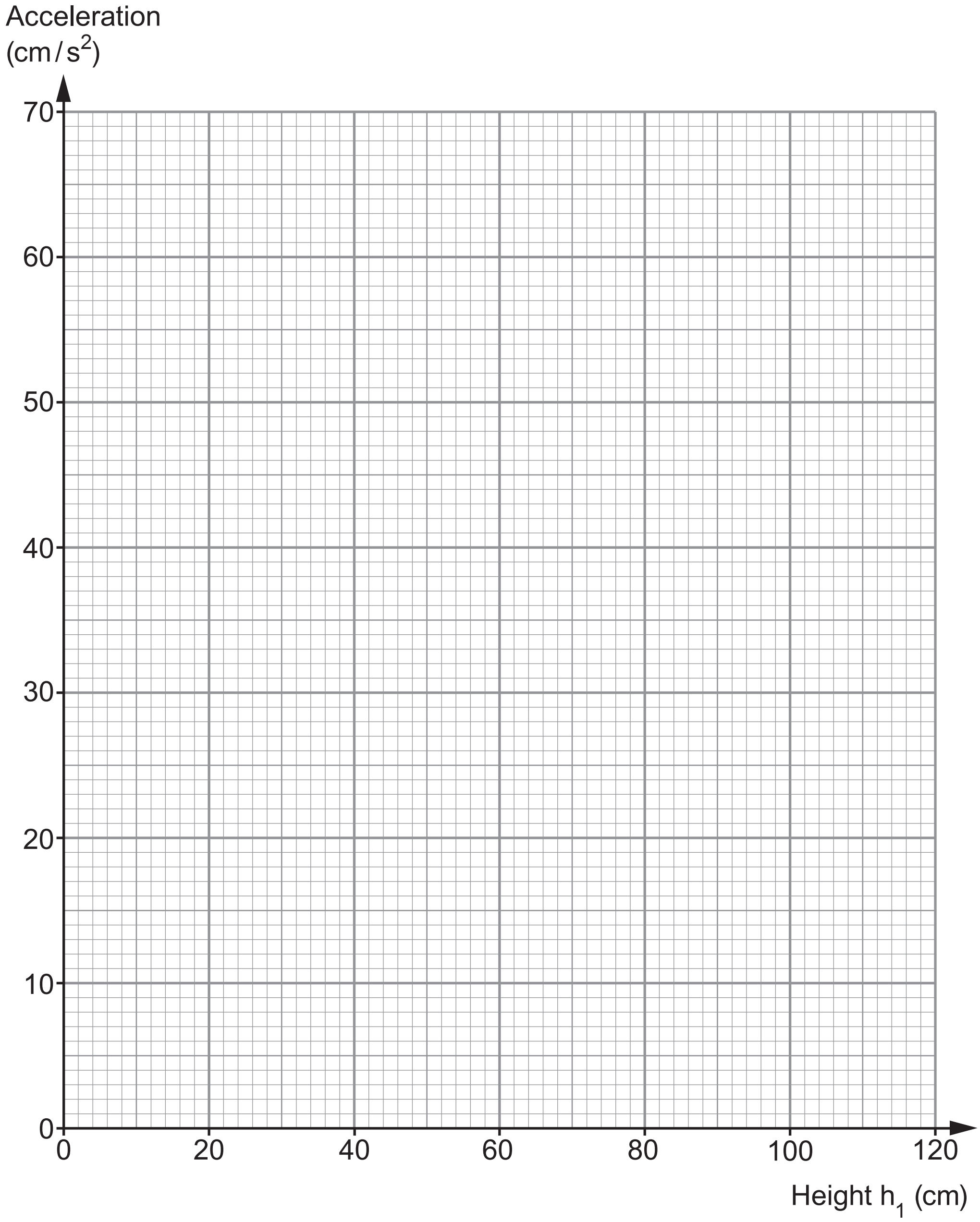
1. ....

2. ....

(ii) The results of the experiment are shown in the table below.

Height h_1 (cm)	Acceleration (cm/s^2)
20	14
40	28
60	40
80	58
100	70

Use the data in the table to plot a graph on the grid opposite and draw a suitable line. [3]



(iii) Use the graph to find the acceleration when the height h_1 is 50 cm. [1]

Acceleration = ... cm/s^2

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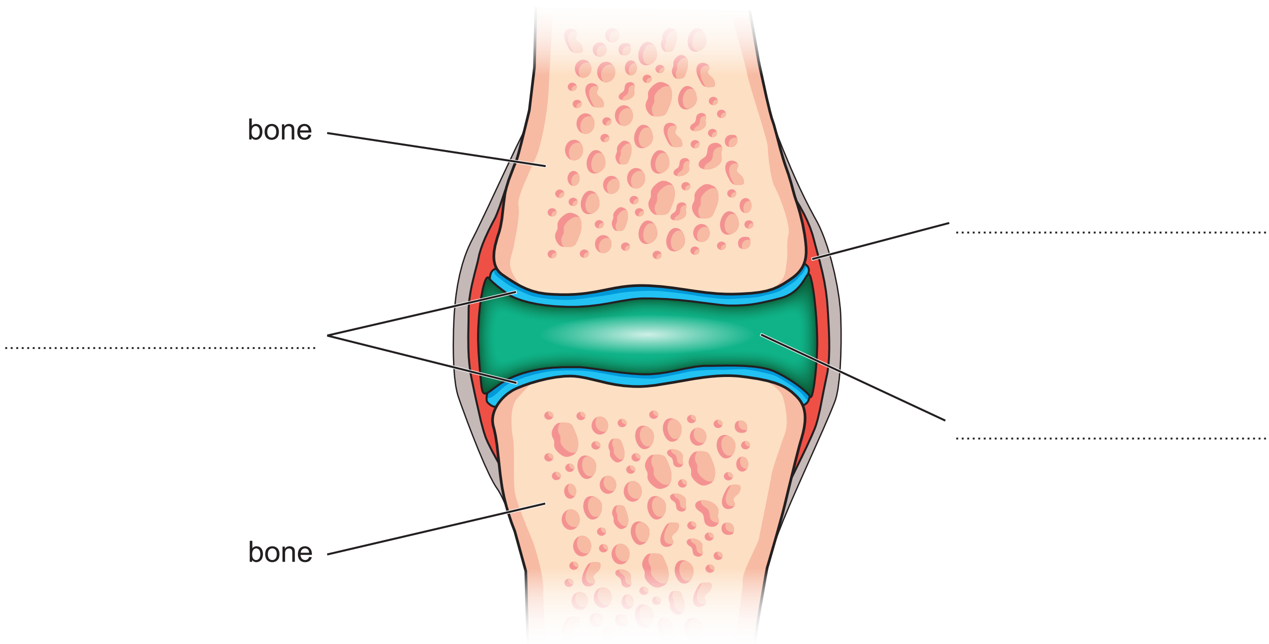
(c) Ski jumpers place their knee joints under considerable strain during their career.

(i) A simplified diagram of a knee joint is shown below. Use words from the box to **complete** the labelling of the diagram. [3]

tendon	ligament	cartilage	synovial fluid	synovial membrane



(ii) An X-ray was taken of a ski jumper's leg following an accident.

Complete the sentence by underlining the term in the brackets. [1]

The X-ray shows a **(simple / compound / green stick)** fracture of the leg.

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